

Customer Credit Risk Dataset

Data Preprocessing & Machine Learning — Theory Notes

Simple English definitions for the topics used in the project.

1. Fundamentals

Data Analysis: Process of examining and understanding data to find useful information and patterns.

Data Science Project: A step-by-step process of collecting, cleaning, analyzing and using data to solve a problem.

Machine Learning Problem Framing: Clearly defining what a Machine Learning model should predict or solve.

Tensor: A structure used to store numbers in one or more dimensions, such as an array or matrix.

NumPy: A Python library used for numerical calculations and arrays.

2. Data Acquisition

Data Acquisition: Collecting data from different sources for analysis.

CSV: A file format that stores table data in rows and columns.

JSON: A structured text format used to store and exchange data.

SQL: A language used to store, manage and retrieve data from databases.

API: A way for one software application to request data from another application or service.

3. Data Understanding

Pandas: A Python library used for loading, cleaning, analyzing and manipulating data.

df.info(): Shows rows, columns, data types and non-null values in a DataFrame.

df.describe(): Shows statistics such as count, mean, standard deviation, minimum and maximum.

Pandas Profiling / YData Profiling: Automatically creates a report with statistics, missing values, distributions, correlations and data-quality warnings.

4. Missing Value Handling

Missing Value: A value that is unavailable or recorded as empty/NaN.

Mean Imputation: Replaces missing numerical values with the column average.

Median Imputation: Replaces missing numerical values with the middle value; useful when outliers exist.

Most Frequent Imputation: Replaces missing categorical values with the category occurring most often.

Missing Indicator: Creates an extra variable showing whether the original value was missing.

Random Sample Imputation: Fills a missing value using a randomly selected observed value from the same column.

KNN Imputer: Fills missing values using values from similar or nearest observations.

MICE: Multiple Imputation by Chained Equations estimates missing values using relationships among variables.

Complete Case Analysis: Removes rows containing missing values and keeps only complete observations.

5. Outlier Handling

Outlier: A value that is unusually high or low compared with most other values.

Z-Score: Shows how far a value is from the mean in standard-deviation units.

IQR: Interquartile Range is Q3 minus Q1 and is commonly used to detect outliers.

Percentile Method: Identifies extreme values using selected lower and upper percentiles.

Winsorization: Limits extreme values by replacing them with selected boundary values.

6. Variables and Dates

Categorical Variable: Contains groups or labels such as gender, region or loan purpose.

Numerical Variable: Contains measurable numbers such as age, income or loan amount.

Ordinal Variable: Contains ordered categories, such as Primary, Graduate and Post-Graduate.

Date Handling: Converts dates into useful features such as year, month and weekday.

7. Encoding

Encoding: Converts categorical information into numerical form for Machine Learning.

Ordinal Encoding: Converts ordered categories into numbers while keeping their order.

Label Encoding: Assigns a numerical label to each category.

One-Hot Encoding: Creates separate binary columns for each category.

Numerical Encoding: Represents categorical information using numerical values.

8. Binning and Binarization

Binning: Converts continuous numerical values into groups or ranges.

Binarization: Converts numerical values into binary values, usually 0 and 1, using a threshold.

Quantile Binning: Creates groups containing a similar number of observations.

K-Means Binning: Uses K-Means clustering to divide values into groups based on similarity.

9. Feature Scaling

Feature Scaling: Changes numerical features to a comparable scale.

Standardization: Uses the mean and standard deviation; values commonly have mean near 0 and standard deviation near 1.

Normalization: Scales data to a common range, often 0 to 1.

Min-Max Scaling: Transforms values to a selected range, commonly 0 to 1.

MaxAbs Scaling: Divides values by the maximum absolute value while preserving the sign.

Robust Scaling: Uses the median and quartiles, so it is less affected by outliers.

10. Transformations

Transformation: Changes the form or distribution of a variable to make it more suitable for analysis or ML.

Log Transformation: Reduces the effect of very large values and can reduce right skewness.

Reciprocal Transformation: Uses $1/\text{value}$ and can change the shape of a skewed distribution.

Square Root Transformation: Uses the square root of values and can reduce moderate skewness.

Box-Cox Transformation: A power transformation mainly used with positive data to improve its distribution.

Yeo-Johnson Transformation: A power transformation that can work with positive and negative values.

11. Feature Construction

Feature Engineering: Creating new useful features from existing data.

Debt-to-Income Ratio: Compares loan/debt amount with income to show debt burden.

Average Monthly Transactions: Represents the average number of transactions per month.

Spending-to-Income Ratio: Compares spending with income to understand spending behavior.

12. Column Transformer and Final Dataset

Column Transformer: Applies different preprocessing methods to different columns in one pipeline.

Preprocessing Pipeline: A sequence of data-preparation steps applied in an organized and repeatable way.

Cleaned Dataset: A dataset where missing values, errors and unwanted problems have been handled.

Transformed Dataset: A dataset whose features have been encoded, scaled or mathematically transformed.

ML-Ready Dataset: A dataset prepared in a suitable format for Machine Learning.

Data Quality: Describes how accurate, complete, consistent and useful data is.

■ Project Flow

Data Acquisition → Data Understanding → Missing Value Handling → Outlier Handling → Encoding → Binning/Binarization → Scaling → Transformation → Feature Engineering → Final Cleaned Dataset → ML Readiness

■ Final Conclusion

The dataset is cleaned and prepared by handling missing values, outliers, categorical variables and numerical features. Scaling, transformations and feature engineering make the final dataset more suitable for Machine Learning.